



Docker One Shot 2026 – Video Notes

This video covers **Docker from absolute basics to real-world backend usage** using **Spring Boot, MySQL, Docker Compose, Volumes, and Networking**.

Video Link : https://youtu.be/kOryO5I_w14



Topics Covered



Docker Fundamentals

- ✓ Why Docker Exists (Problem First Approach)
 - ✓ What is Docker?
 - ✓ Docker Architecture Explained
 - ✓ Docker Installation & Setup
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Containerizing Spring Boot Application

- ✓ Creating Dockerfile
 - ✓ Port Mapping Explained
 - ✓ Environment Variable Configuration
 - ✓ Running Spring Boot inside Docker
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Docker Core Concepts

- ✓ Docker Images vs Containers

- ✓ Docker Volumes (Data Persistence)
 - ✓ Docker Networking (Container Communication)
 - ✓ Docker Compose (Multi-container setup)
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What is Docker?

Docker is a platform that allows you to package your **application**, **runtime**, **libraries**, and **dependencies** into a single unit called a **container**.

This ensures your application runs the same way in:

- local machine
- testing environment
- staging
- production

👉 *“Works on my machine” problem — solved.*

What is a Dockerfile?

A **Dockerfile** is a text file that contains **step-by-step instructions** used to build a Docker image.

It acts as a **blueprint** for creating containers.

Common Dockerfile Instructions

1. **Base Image** – Operating system + runtime
2. **WORKDIR** – Working directory inside container

3. **COPY** – Copy application files
 4. **EXPOSE** – Application port
 5. **ENTRYPOINT / CMD** – Startup command
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What is Docker Compose?

Docker Compose allows us to **define and run multi-container Docker applications** using a simple **YAML file**.

With Docker Compose, we can:

- run multiple containers together
- create networks automatically
- manage volumes easily
- define environment variables
- start everything using one command

`docker compose up`

Real-World Example Used

- Spring Boot application
- MySQL database
- Docker volumes for persistence
- Docker networking using service names
- Environment-based configuration

- Fully containerized backend system